

Determinants

ANSWER KEY



2x2 determinants

- 1 Find the determinant of the matrix with rows $\langle 4, 3 \rangle, \langle 2, 5 \rangle$.
 $4(5) - 3(2) = 20 - 6 = 14$.
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3x3 determinants via cofactor expansion

- 2 Find the determinant of the matrix with rows $\langle 1, 2, 3 \rangle, \langle 0, 1, 4 \rangle, \langle 5, 6, 0 \rangle$ by cofactor expansion along the first row.
 $1\det(\langle 1, 4 \rangle, \langle 6, 0 \rangle) - 2\det(\langle 0, 4 \rangle, \langle 5, 0 \rangle) + 3\det(\langle 0, 1 \rangle, \langle 5, 6 \rangle) = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1$
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The determinant product rule

- 3 For A with rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B with rows $\langle 2, 1 \rangle, \langle 0, 3 \rangle$, verify $\det(AB) = \det(A)\det(B)$.
 $\det A = 4 - 6 = -2$, $\det B = 6 - 0 = 6$, so $\det A \det B = -12$. AB has rows $\langle 2, 7 \rangle, \langle 6, 15 \rangle$ (computed as usual matrix multiplication), and $\det(AB) = 2(15) - 7(6) = 30 - 42 = -12$. They match.
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The invertibility criterion

- 4 State the invertibility criterion in terms of the determinant.
A square matrix A is invertible if and only if $\det A \neq 0$.
- 5 Is the matrix with rows $\langle 2, 4 \rangle, \langle 1, 2 \rangle$ invertible? Justify using the determinant.
 $\det = 2(2) - 4(1) = 0$. Since the determinant is 0, the matrix is NOT invertible (singular).
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Applying the invertibility criterion

- 6 For what value(s) of k is the matrix with rows $\langle k, 2 \rangle, \langle 3, k \rangle$ NOT invertible?
 $\det = k^2 - 6$. Setting $k^2 - 6 = 0$ gives $k = \pm\sqrt{6}$. The matrix fails to be invertible exactly when $k = \sqrt{6}$ or $k = -\sqrt{6}$.